Linear Models: CEF and Best Linear Predictor

Robert Gulotty

University of Chicago

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Populations, Projections, and Structure

- Today we study relationships between random variables Y and $X = (X_1, \ldots, X_k)$ in the population.
- The **Conditional Expectation Function (CEF)**: $m(X) = \mathbb{E}[Y \mid X]$.
  A potentially non-linear expectation of Y based on X.
- The **Best Linear Predictor (BLP)**: $\beta^* = \arg\min_b \mathbb{E}[(Y - X'b)^2]$.
  The linear projection of Y onto X in the population.
- Later we will use algebra to fit a line in the sample and use probability theory to estimate the BLP.
- Structural (exogeneity) assumptions determine whether the BLP coincides with the CEF.

Properties of Conditional Expectation

- $m(X) \equiv \mathbb{E}[Y \mid X]$ (Definition)
  * Note: $\mathbb{E}[Y|X]$ is a random variable (a function of X), not a number!
- $\mathbb{E}[\mathbb{E}[Y \mid X] \mid X] = \mathbb{E}[Y \mid X]$ (Idempotence)
- $\mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[Y]$ (Law of Iterated Expectations)
- $m(X) = \arg\min_g \mathbb{E}[(Y - g(X))^2]$ (Best predictor of Y using X)

CEF Error and Mean Independence

- Define the CEF error: $e \equiv Y - m(X)$, so

$$Y = Y + (m(X) - m(X)) = m(X) + (Y - m(X)) = m(X) + e$$

- The CEF error has conditional mean zero (by construction):

$$\mathbb{E}[e|X] = \mathbb{E}[Y - m(X)|X] = m(X) - m(X) = 0$$

- And unconditional mean zero (by LIE):

$$\mathbb{E}[e] = \mathbb{E}[\mathbb{E}[e|X]] = 0$$

- $\mathbb{E}[e|X] = 0$ is called *mean independence*.

Hansen Theorem 2.4: Properties of the CEF Error e

1. $\mathbb{E}[e|X] = 0$
2. $\mathbb{E}[e] = 0$
3. If the rth moment of Y exists, the rth moment of e exists. [Technical]
4. For *any measurable function* h(X) where the expected product with e exists $[\mathbb{E}[|h(X)e|] < \infty]$, then, $\mathbb{E}[h(X)e] = 0$

Key Point

These are properties, not assumptions; they follow from LIE and the definition of the CEF: these properties are why we use the CEF as a mathematical tool.

Why Covariance Matters for Decomposition

Suppose we decompose a random variable into two parts: $Y = S + N$ ("signal" + "noise").

The variance of the sum is:

$$\text{Var}(Y) = \text{Var}(S) + \text{Var}(N) + 2\,\text{Cov}(S, N)$$

- If $\text{Cov}(S, N) \neq 0$, there is no clean separation: signal and noise share variation, and we cannot attribute variance to one or the other.
- If $\text{Cov}(S, N) = 0$, we get a **clean decomposition**:

$$\text{Var}(Y) = \text{Var}(S) + \text{Var}(N)$$

Question: Is there a function of X that decomposes Y into uncorrelated signal and noise?

Covariance of CEF and Its Error

Yes—the CEF does exactly this. Since e is a random variable, compute its covariance with $m(X)$:

$$\mathbb{E}[e\, m(X)] = \mathbb{E}[\mathbb{E}[e\, m(X)|X]] = \mathbb{E}[m(X)\, \mathbb{E}[e|X]] = \mathbb{E}[m(X) \cdot 0] = 0$$

Because $m(X)$ is a measurable function of X, Theorem 2.4.4 applies.

$$\text{Cov}(e, m(X)) = \mathbb{E}[e\, m(X)] - \mathbb{E}[e]\, \mathbb{E}[m(X)] = 0 - 0 \cdot \mathbb{E}[m(X)] = 0$$

So the **CEF decomposition** gives us exactly the clean split we wanted:

$$\text{Var}(Y) = \text{Var}(m(X)) + \text{Var}(e)$$

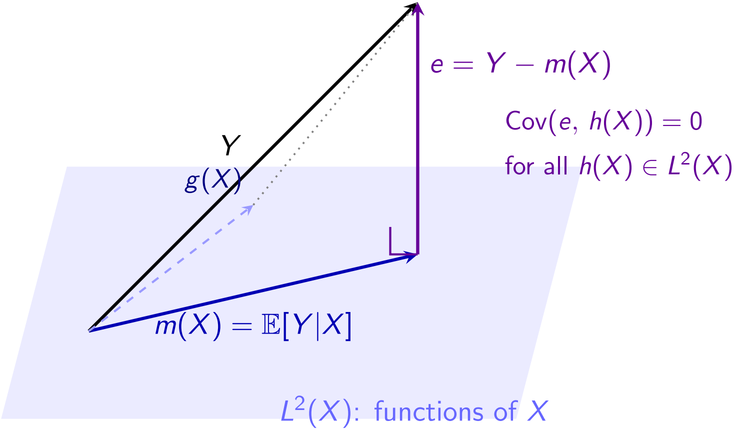
Why Orthogonality? The Projection Interpretation

This is not a mathematical coincidence. It follows from what the CEF *is*:

- The CEF minimizes $\mathbb{E}[(Y - g(X))^2]$ over **all** functions of X (Thm 2.7).
- In the language of Hilbert spaces, $m(X)$ is the **orthogonal projection** of Y onto the space of all square-integrable functions of X.
- A defining property of orthogonal projection: the residual is perpendicular to the space you project onto.
- Since $m(X)$, X, X^2, $\sin(X)$, $\ldots$ are all in that space, the error e must be uncorrelated with *every* function of X.

The CEF decomposition $\text{Var}(Y) = \text{Var}(m(X)) + \text{Var}(e)$ is a Pythagorean theorem: it holds because signal and noise are orthogonal.

Visualizing Projection in L^2 Space



The CEF is the closest point in $L^2(X)$ to Y. The residual e is orthogonal to the subspace.

A Note on the Geometry

The diagram on the previous slide is an *analogy*, not a literal picture. Random variables do not live in $\mathbb{R}^3$—they live in an abstract **Hilbert space** L^2.

Euclidean $\mathbb{R}^n$		**Hilbert space** L^2
Vector $\mathbf{v}$	$\longleftrightarrow$	Random variable Y
Dot product $\mathbf{u} \cdot \mathbf{v}$	$\longleftrightarrow$	$\mathbb{E}[UV]$ (inner product)
Length $\|\mathbf{v}\|^2$	$\longleftrightarrow$	$\mathbb{E}[Y^2]$ (second moment)
Perpendicularity $\mathbf{u} \cdot \mathbf{v} = 0$	$\longleftrightarrow$	Orthogonality $\mathbb{E}[UV] = 0$
Projection onto subspace	$\longleftrightarrow$	Conditional expectation

You do *not* need to know Hilbert space theory for this course. The takeaway is just the intuition: projection $\Rightarrow$ orthogonal residual $\Rightarrow$ clean variance decomposition.

How Much Does the CEF Leave Unexplained?

The CEF $m(X)$ captures the systematic part of Y. The error $e = Y - m(X)$ is everything else.

Unconditional error variance:

$$\sigma^2 \equiv \text{Var}[e] = \mathbb{E}[e^2]$$

The average prediction error across all values of X.

Conditional error variance:

$$\sigma^2(x) = \text{Var}[e|X = x] = \mathbb{E}[e^2|X = x]$$

How uncertain is our prediction *for a specific* $X = x$? Averaging: $\sigma^2 = \mathbb{E}[\sigma^2(X)]$.

Example: Predict vote share from incumbency. $\sigma^2(D{=}1)$ may differ from $\sigma^2(D{=}0)$—incumbents might have more predictable outcomes than challengers.

Mean-Variance Representation

Define $u = \frac{e}{\sigma(X)}$, so $\mathbb{E}[u|X] = 0$ and $\text{Var}[u|X] = 1$. Then:

$$Y = m(X) + \sigma(X)\, u$$

X matters in **two ways**:

- Through the **mean**: $m(X)$ shifts the expected outcome.
- Through the **variance**: $\sigma(X)$ scales the noise.

If $\sigma(X)$ is constant across all X: **homoskedastic** errors.
If $\sigma(X)$ varies with X: **heteroskedastic** errors.

Example: In a model of household income, $\sigma(\text{Education})$ tends to increase with education—PhDs can be professors or hedge fund managers, while high-school graduates have a narrower income range.

Hansen Theorem 2.6

Theorem 2.6 (More Variables Reduce Error Variance)

Let $e_1 = Y - m(X_1)$ and $e_{12} = Y - m(X_1, X_2)$. Then:

$$\text{Var}[Y] \geq \text{Var}[e_1] \geq \text{Var}[e_{12}]$$

You will always explain (weakly) more of the variance with more variables. This is why R^2 is a poor measure of model fit.

Proof strategy: Show $\text{Var}[m(X_1)] \leq \text{Var}[m(X_1, X_2)]$—the finer CEF has (weakly) larger variance. Then the **CEF decomposition** $\text{Var}[Y] = \text{Var}[m(X)] + \mathbb{E}[\text{Var}[Y|X]]$ converts this into the result about error variances.

Proof of Theorem 2.6: Setup

Goal: Show $\mathbb{E}[m(X_1)^2] \leq \mathbb{E}[m(X_1, X_2)^2]$, where $m(\cdot)$ denotes the relevant CEF.

Step 1. By LIE, the coarser CEF is a conditional expectation of the finer one:

$$m(X_1) \;=\; \mathbb{E}[Y|X_1] \;=\; \mathbb{E}\big[\mathbb{E}[Y|X_1, X_2] \;\big|\; X_1\big] \;=\; \mathbb{E}\big[\, m(X_1, X_2) \;\big|\; X_1\big]$$

Step 2. Jensen's inequality: for any convex function φ (such as $z \mapsto z^2$),

$$\varphi(\mathbb{E}[Z|X_1]) \;\leq\; \mathbb{E}\big[\varphi(Z) \;\big|\; X_1\big]$$

Apply with $Z = m(X_1, X_2)$ and $\varphi(z) = z^2$:

$$\Big(\mathbb{E}\big[m(X_1, X_2) \;\big|\; X_1\big]\Big)^2 \;\leq\; \mathbb{E}\big[m(X_1, X_2)^2 \;\big|\; X_1\big]$$

The left side is $m(X_1)^2$ by Step 1, so:

$$m(X_1)^2 \;\leq\; \mathbb{E}\big[m(X_1, X_2)^2 \;\big|\; X_1\big]$$

Proof of Theorem 2.6: Conclusion

Step 3. Take unconditional expectations of both sides:

$$\mathbb{E}\big[m(X_1)^2\big] \;\leq\; \mathbb{E}\big[m(X_1, X_2)^2\big]$$

Step 4. Convert to variances. Recall $\text{Var}[Z] = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2$, and by LIE:

$$\mathbb{E}[m(X_1)] = \mathbb{E}[m(X_1, X_2)] = \mathbb{E}[Y]$$

so $(\mathbb{E}[Z])^2$ is the same for both. Subtracting it from the inequality above:

$$\text{Var}[m(X_1)] \;\leq\; \text{Var}[m(X_1, X_2)]$$

Step 5. Since $\text{Var}[Y] = \text{Var}[m] + \text{Var}[e]$ (by the CEF decomposition), larger $\text{Var}[m]$ means smaller $\text{Var}[e]$:

$$\text{Var}[e_1] \;\geq\; \text{Var}[e_{12}] \qquad \square$$

Hansen Theorem 2.7: CEF as Best Predictor

Theorem 2.7 (CEF Is the Best Predictor)

If $\mathbb{E}[Y^2] < \infty$, then for any measurable function $g(X)$: $\mathbb{E}\left[(Y - m(X))^2\right] \leq \mathbb{E}\left[(Y - g(X))^2\right]$. The CEF $m(X) = \mathbb{E}[Y|X]$ minimizes mean squared prediction error.

Proof:

$$
\begin{aligned}
\mathbb{E}\left[(Y - g(X))^2\right] &= \mathbb{E}\left[(e + m(X) - g(X))^2\right] && (Y = e + m(X)) \\
&= \mathbb{E}[e^2] + 2\mathbb{E}[e(m(X) - g(X))] + \mathbb{E}[(m(X) - g(X))^2] \\
&= \mathbb{E}[e^2] + 0 + \mathbb{E}[(m(X) - g(X))^2] && \text{(by thm 2.4.4)} \\
&\geq \mathbb{E}[e^2] = \mathbb{E}[(Y - m(X))^2] && \text{(squares} \geq 0)
\end{aligned}
$$

Implications of Theorem 2.7

- Regardless of the distribution of X and Y, the best predictor is the CEF $m(X)$.
  - For example: Consider the intercept only model $\mu = \mathbb{E}[Y]$.
  - The best predictor for Y, among constants, is μ.
- Since the CEF is the best predictor, anything else must be identical or worse.
- We don't know what the joint distribution is, so we don't generally know the CEF, we have to use an estimator.

Linear CEF and Best Prediction

If $m(X)$ is *linear* in $X = (X_1, \ldots, X_k, 1)'$, we can write $m(X) = X'\beta$. The **linear regression model** is:

$$Y = X'\beta + e, \quad \mathbb{E}[e|X] = 0$$

If homoskedastic: $\mathbb{E}[e^2|X] = \sigma^2$.

Best prediction property:

- If the CEF is linear, the solution to $\beta = \arg\min_b \mathbb{E}[(Y - X'b)^2]$ gives the right formula for β.
- If the CEF is nonlinear, the solution gives the best linear approximation to the true CEF.

From CEF to Linear Predictors

- The CEF provides: summarization, MMSE prediction, and decomposition.
- But the CEF requires knowing the conditional distribution $f_{Y|x}(y|x)$.
  - Exception: CEF is always linear if X is discrete and all interactions are included.
- We will instead approximate the CEF using a **Best Linear Predictor** (BLP).
- Later we will show that OLS estimates are optimal **if** the CEF is linear and the residuals are homoskedastic.

The CEF in Political Science

- **Voter turnout and age:** $\mathbb{E}[\text{Turnout}|\text{Age}]$ is nonlinear—turnout rises steeply from 18–30, plateaus in middle age, and rises again among retirees.
- **Income and party ID:** $\mathbb{E}[\text{Income}|\text{Party}]$ differs across categories—the CEF is just group means when X is discrete.
- **Conflict and GDP:** $\mathbb{E}[\text{Conflict}|\text{GDP}]$ may be highly nonlinear, with sharp thresholds at low GDP.

Takeaway: The CEF is the target. But when the relationship is complex, we approximate it with a linear predictor and interpret accordingly.

What Have We Learned About the CEF?

CEF Summary

1. $m(X) = \mathbb{E}[Y|X]$ is the **best predictor** of Y given X (Thm 2.7).
2. The CEF error $e = Y - m(X)$ satisfies $\mathbb{E}[e|X] = 0$ **by construction**—not an assumption (Thm 2.4).
3. Signal and noise are uncorrelated: $\text{Cov}(m(X), e) = 0$.
4. More variables $\Rightarrow$ (weakly) smaller error variance (Thm 2.6).
5. $Y = m(X) + \sigma(X)u$: X shapes both the mean *and* the variance.

Next: We approximate $m(X)$ with a *linear* function—the Best Linear Predictor.

(Best) Linear Predictors (bivariate case)

- Suppose we want a predictor $\mathbb{E}^*(Y|X)$ that is **linear**:

$$f(X) = a + bX$$

  Our standard for prediction is to minimize the mean-square error (best):

- Whereas the CEF might be infinitely complex, the BLP is characterized just by two numbers, a and b.
  Choose a and b to minimize

$$M = \mathbb{E}[(Y - (a + bX))^2]$$

Solving for Best Linear Predictor (a and b)

FOC w.r.t. a:

$$0 = -2\mathbb{E}[Y] + 2\mathbb{E}[a + bX] \implies a = \mathbb{E}[Y] - b\mathbb{E}[X]$$

FOC w.r.t. b:

$$\begin{aligned}
0 &= -2\mathbb{E}[YX] + 2\mathbb{E}[(a + bX)(X)] \\
\mathbb{E}[YX] &= [\mathbb{E}[Y] - b\mathbb{E}[X]]\mathbb{E}[X] + b\mathbb{E}[X^2] \\
\mathbb{E}[YX] - \mathbb{E}[Y]\mathbb{E}[X] &= b[\mathbb{E}[X^2] - \mathbb{E}[X]^2] \\
b &= \frac{Cov(X, Y)}{Var(X)}
\end{aligned}$$

Best Linear Predictor

The best linear predictor is the population linear projection:

$$\mathbb{E}^*[Y|X] = \beta_0 + \beta_1 X$$
$$Y = \beta_0 + \beta_1 X + e$$

- Parameters (in Greek)
  - The slope $\beta = \frac{Cov(X,Y)}{Var(X)}$
  - The intercept $\alpha = \mathbb{E}(Y) - \beta\mathbb{E}(X)$
- $\mathcal{P}(Y|X) = \mathbb{E}[Y] + \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)} \cdot (X - \mathbb{E}[X])$

Deriving Variance of BLP at Minimized Values

$$\beta = \frac{Cov(X,Y)}{Var(X)} \qquad \beta Var(X) = Cov(X,Y)$$

$$\begin{aligned}
\mathbb{E}[(Y-(\alpha+\beta X))^2] - \mathbb{E}[Y-(\alpha+\beta X)]^2 &= Var(Y-(\alpha+\beta X)) \\
&= Var(Y-\beta X) \\
&= Var(Y) + \beta^2 Var(X) - 2\beta Cov(X,Y) \\
&= Var(Y) + \beta^2 Var(X) - 2\beta^2 Var(X) \\
&= Var(Y) - \beta^2 Var(X) \\
\sigma_\epsilon^2 &= \sigma_Y^2 - \beta^2 \sigma_X^2
\end{aligned}$$

Example: Population Linear Projection is not the CEF

Example: $Y = X + X^2$, $X \sim N(0,1)$, $\mathbb{E}[X] = \mathbb{E}[X^3] = 0$, $\mathbb{E}[X^2] = 1$
True CEF: $m(X) = X + X^2$
Population Linear Projection: $Y = \alpha + \beta X + e$.

$$
\begin{aligned}
\alpha &= \mathbb{E}[Y] - \beta \mathbb{E}[X] = \mathbb{E}[X + X^2] - \beta \cdot 0 = 0 + 1 - 0 \\
\beta &= \text{Cov}(X, Y)/\text{Var}(X) = \text{Cov}(X, X + X^2)/1 = 1/1 \\
\mathcal{P}(Y|X) &= 1 + 1 * X \\
e &= m(X) - \mathcal{P}(Y|X) = X^2 - 1 \\
\mathbb{E}[eX] &= \mathbb{E}[(X^2 - 1)X] = \mathbb{E}[X^3 - X] = 0
\end{aligned}
$$

The BLP error $e = X^2 - 1$ is a function of X, so $\mathbb{E}[e|X] = X^2 - 1 \neq 0$.

$\mathbb{E}[Xe] = 0$ holds, but $\mathbb{E}[e|X] = 0$ fails—the error has a systematic nonlinear pattern in X. This is what "weaker" means: $\mathbb{E}[e|X] = 0 \implies \mathbb{E}[Xe] = 0$, but not vice versa.

CEF vs Linear Predictors / Projection

- CEF ($m(X)$)
  - Best prediction, the prediction that minimizes squared error.
  - Can be non-linear.
  - Requires knowing joint distribution.
  - Prediction Errors are uncorrelated with any function of X. ($E(e|X) = 0$.)
- Linear Predictor/ Projection ($\mathcal{P}(Y|X)$)
  - Draws a line (plane/hyperplane)
  - Requires knowing variances and covariances.
  - Prediction Errors are uncorrelated with X ($E(Xe) = 0$.)

CEF vs BLP: The Big Picture

Why Settle for Linear?

- The CEF is the *best possible* predictor—but we rarely know it.
- The BLP is the best *linear* predictor—and we only need $\text{Cov}(X, Y)$ and $\text{Var}(X)$.
- When the CEF is actually linear (e.g., jointly normal X, Y), BLP = CEF.
- When the CEF is nonlinear, BLP gives the best linear approximation.

Key insight: $\mathbb{E}[Xe] = 0$ (BLP errors uncorrelated with X) is *weaker* than $\mathbb{E}[e|X] = 0$ (CEF errors mean-independent of X). The BLP projection "uses up" the linear information in X, but may leave nonlinear patterns in the residual.

Why Write $\mathbb{E}[Xe] = 0$?

- Saying "X is exogenous" is intuitive but vague. Writing $\mathbb{E}[Xe] = 0$ buys you three things:
  1. **It makes assumptions precise.** "Exogenous" could mean many things. $\mathbb{E}[Xe] = 0$ says *exactly* what must hold: the prediction error $e = Y - X'\beta$ is uncorrelated with X.
  2. **It generalizes.** Swap X for an instrument Z and you get IV: $\mathbb{E}[Ze] = 0$. Add more conditions and you get GMM. Every estimator in this course is defined by *which* moment conditions it solves.
  3. **It defines the parameter.** The population β is the value that makes $\mathbb{E}[X(Y - X'\beta)] = 0$. The OLS formula $(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{Y}$ solves the sample analog of this condition.

From Theory to Moment Conditions: A Preview

Every empirical model in this course follows the same logic:

1. **Substantive claim:** A mechanism, institution, or design feature creates a restriction on the joint distribution.
2. **Moment condition:** Translate that claim into $\mathbb{E}[g(W,\theta)] = 0$.
3. **Estimator:** Find $\hat{\theta}$ that solves the sample analog.

Example: In a randomized experiment, random assignment means potential outcomes are independent of treatment. This implies $\mathbb{E}[De] = 0$, and OLS consistently estimates the ATE.

Counterexample: If subjects self-select into treatment based on expected benefits, $\mathbb{E}[De] \neq 0$. OLS is inconsistent; you need a *different* moment condition (e.g., from an instrument). **The question is always:** Why should this moment condition hold?

BLP in R: Bivariate Case

```r
# Population formula: beta = Cov(X,Y) / Var(X)
beta_formula <- cov(x, y) / var(x)
alpha_formula <- mean(y) - beta_formula * mean(x)

# OLS estimates the same thing from a sample
mod <- lm(y ~ x)
coef(mod)  # compare to c(alpha_formula, beta_formula)
```

From Bivariate to Multivariate Linear Prediction

- So far we derived the BLP for a single regressor: $\beta = \text{Cov}(X, Y)/\text{Var}(X)$.
- In practice, we want to predict Y using *many* variables: $X = (X_1, X_2, \ldots, X_k, 1)'$.
- The algebra generalizes: scalar variances and covariances become **matrices**.

Definition (Mean Square Prediction Error)

The **MSPE** *of a linear predictor $X'\beta$ is:*

$$S(\beta) = \mathbb{E}\left[(Y - X'\beta)^2\right]$$

This is the expected squared distance between Y and our prediction. The **best linear predictor** *minimizes $S(\beta)$ over all $\beta \in \mathbb{R}^k$.*

Regularity Conditions

- Given Y and $X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ 1 \end{pmatrix}$, we need three conditions for the BLP to exist:

   1. $\mathbb{E}[Y^2] < \infty$ (finite second moment of Y)
   2. $\mathbb{E}\|X\|^2 < \infty$ (finite second moments of regressors)
   3. $\mathbf{Q}_{XX} = \mathbb{E}[XX']$ is positive definite (no perfect collinearity)

- Conditions 1–2 ensure $S(\beta)$ is finite. Condition 3 ensures a *unique* minimizer.
- If $\mathbf{Q}_{XX}$ is not positive definite, some linear combination of regressors is constant—and the coefficients cannot be separately identified.

Deriving the Linear Projection Coefficient β

$$\begin{aligned}
S(\beta) &= \mathbb{E}[(Y - X'\beta)(Y - X'\beta)] \\
&= \mathbb{E}[Y^2] - 2\beta'\mathbb{E}[XY] + \beta'\mathbb{E}[XX']\beta \\
0 &= \frac{\partial}{\partial\beta} S(\beta) = -2\mathbb{E}[XY] + 2\mathbb{E}[XX']\beta \\
\mathbf{Q}_{XY} &= \mathbf{Q}_{XX}\beta \\
\mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY} &= \beta \\
\mathbb{E}[XX']^{-1}\mathbb{E}[XY] &= \beta
\end{aligned}$$

Linear Projection and Projection Error

$$\mathcal{P}[Y|X] = X'\beta = X'\mathbb{E}[XX']^{-1}\mathbb{E}[XY]$$

The projection error is $e = Y - X'\beta$, so $Y = X'\beta + e$.

The error is uncorrelated with X:

$$\mathbb{E}[Xe] = \mathbb{E}[XY] - \mathbb{E}[XX']\mathbb{E}[XX']^{-1}\mathbb{E}[XY] = 0$$

$$\begin{pmatrix} \mathbb{E}[X_1 e] \\ \mathbb{E}[X_2 e] \\ \vdots \\ \mathbb{E}[1 * e] \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

This is k equations, the last of which is $\mathbb{E}[e] = 0$ when there is a constant.

What If There Is No Intercept?

If X does not include a constant, the moment conditions are only $\mathbb{E}[X_j e] = 0$ for each regressor—there is no equation forcing $\mathbb{E}[e] = 0$.

Consequences:

- The projection line is forced through the origin: $\mathcal{P}[Y|X] = X'\beta$ with no intercept.
- The error e can have a nonzero mean: $\mathbb{E}[e] \neq 0$ in general.
- The usual decomposition $\text{Var}(Y) = \text{Var}(X'\beta) + \text{Var}(e)$ still holds (since $\mathbb{E}[Xe] = 0$), but $\mathbb{E}[Y] \neq \mathbb{E}[X'\beta]$.

Takeaway: Always include a constant unless you have a substantive reason not to. Omitting it forces the regression through the origin and allows systematic over- or under-prediction on average.

The Design Matrix

$\beta = \mathbb{E}[XX']^{-1}\mathbb{E}[XY]$ exists and is unique *if* $\mathbf{Q}_{XX} = \mathbb{E}[XX']$ is invertible.

$\mathbf{Q}_{XX}$ is the population second moment matrix of X.

For any non-zero $\alpha \in \mathbb{R}^k$

$$\alpha' \mathbf{Q}_{XX} \alpha = \mathbb{E}[\alpha' XX' \alpha] = \mathbb{E}[(\alpha' X)^2] \geq 0$$

This must be strictly > 0 to be invertible, if not, there is no unique solution to $\mathbf{Q}_{XY} = \mathbf{Q}_{XX}\beta$, and we say β is not identified.

BLP in R: Multiple Regressors

Now that we have the matrix formula $\beta = \mathbb{E}[XX']^{-1}\mathbb{E}[XY]$, we can compute it directly:

```r
# Build the design matrix (include a column of 1s)
X <- cbind(1, x1, x2)

# Solve (X'X) beta = X'y directly, without inverting
beta_matrix <- solve(t(X) %*% X, t(X) %*% y)

# lm() gives the same coefficients, but uses QR
coef(lm(y ~ x1 + x2))
```

`solve(A, b)` solves $Ax = b$ via LU factorization without computing A^{-1}, which is more stable than `solve(A) %*% b`. But `lm()` is even better: it uses QR decomposition on X directly, avoiding the formation of $X'X$ entirely (which squares the condition number).

Application: Linear Probability Model (Binary Y)

$$P(Y = 1|X) = \beta_0 + \beta_1 X$$

- Suppose we have a binary outcome: $Y = \mathbb{I}(\omega \in B)$
- β_0 is the probability that $Y = 1$ if $X = 0$
- β_1 is the change in probability that $Y = 1$ given a one unit change in X.
- The BLP is still the best linear predictor: $\beta = \mathbb{E}[XX']^{-1}\mathbb{E}[XY]$.
- However, it is an approximation to a probability, not itself a probability.

Linear Predictor Error Variance

- Hansen defines $\sigma^2 = \mathbb{E}[e^2]$, $Q_{YY} = \mathbb{E}[Y^2]$ and $\mathbf{Q}_{YX} = \mathbb{E}[YX']$, a $1 \times k$ vector.
- Recall, X and β are $k \times 1$ vectors.

$$
\begin{aligned}
\sigma^2 &= \mathbb{E}[(Y - X'\beta)^2] \\
&= \mathbb{E}[Y^2] - 2\mathbb{E}[YX']\beta + \beta'\mathbb{E}[XX']\beta \\
&= Q_{YY} - 2\boldsymbol{Q}_{YX}\boldsymbol{Q}_{XX}^{-1}\boldsymbol{Q}_{XY} + \boldsymbol{Q}_{YX}\boldsymbol{Q}_{XX}^{-1}\boldsymbol{Q}_{XX}\boldsymbol{Q}_{XX}^{-1}\boldsymbol{Q}_{XY} \\
&= Q_{YY} - \boldsymbol{Q}_{YX}\boldsymbol{Q}_{XX}^{-1}\boldsymbol{Q}_{XY} \\
&\equiv Q_{YY\cdot X}
\end{aligned}
$$

Intercepts and Slopes

- We will often exclude the 1 from X and write the linear projection as:

$$Y = X'\beta + \alpha + e$$

- Demeaning we have a useful reinterpretation of the projection coefficient:

$$\begin{aligned} Y - \mu_Y &= (X - \mu_X)'\beta + e \\ \beta &= (\mathbb{E}[(X - \mu_X)(X - \mu_X)'])^{-1}\mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \\ \beta &= var[X]^{-1}cov(X, Y) \end{aligned}$$

Partitions and Regression Sub-Vectors

- Divide our regressors into $X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$, so the linear projection is:

$$Y = X_1'\beta_1 + X_2'\beta_2 + e, \quad \mathbb{E}[Xe] = 0$$

- We can partition $\mathbf{Q}_{XX}$ and $\mathbf{Q}_{XY}$:

$$\mathbf{Q}_{XX} = \begin{bmatrix} \mathbf{Q}_{11} & \mathbf{Q}_{12} \\ \mathbf{Q}_{21} & \mathbf{Q}_{22} \end{bmatrix} \quad \mathbf{Q}_{XY} = \begin{bmatrix} \mathbf{Q}_{1Y} \\ \mathbf{Q}_{2Y} \end{bmatrix}$$

A Useful Formula

- Recall the error variance was: $Q_{YY \cdot X} \equiv Q_{YY} - \boldsymbol{Q}_{YX}\boldsymbol{Q}_{XX}^{-1}\boldsymbol{Q}_{XY}$
- Analogously $\mathbf{Q}_{11 \cdot 2} \equiv \mathbf{Q}_{11} - \boldsymbol{Q}_{12}\boldsymbol{Q}_{22}^{-1}\boldsymbol{Q}_{21}$,and $\mathbf{Q}_{22 \cdot 1} \equiv \mathbf{Q}_{22} - \boldsymbol{Q}_{21}\boldsymbol{Q}_{11}^{-1}\boldsymbol{Q}_{12}$, etc.
- In each case, $\mathbf{Q}_{a \cdot b}$ is the variation in a not predicted by the linear projection of b.

Solving the Partitioned Normal Equations

The normal equation $\mathbf{Q}_{XY} = \mathbf{Q}_{XX}\beta$ expands into two block rows:

$$\mathbf{Q}_{1Y} = \mathbf{Q}_{11}\beta_1 + \mathbf{Q}_{12}\beta_2 \tag{Row 1}$$

$$\mathbf{Q}_{2Y} = \mathbf{Q}_{21}\beta_1 + \mathbf{Q}_{22}\beta_2 \tag{Row 2}$$

Goal: Eliminate β_2 to isolate β_1.

Step 1. Solve Row 2 for β_2:

$$\beta_2 = \mathbf{Q}_{22}^{-1}\left(\mathbf{Q}_{2Y} - \mathbf{Q}_{21}\beta_1\right)$$

Solving the Partitioned Normal Equations (cont.)

Step 2. Substitute β_2 into Row 1:

$$\mathbf{Q}_{1Y} = \mathbf{Q}_{11}\beta_1 + \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\left(\mathbf{Q}_{2Y} - \mathbf{Q}_{21}\beta_1\right)$$
$$\mathbf{Q}_{1Y} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{2Y} = \left(\mathbf{Q}_{11} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21}\right)\beta_1$$

Step 3. Recognize the residualized quantities from the previous slide:

$$\underbrace{\mathbf{Q}_{1Y} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{2Y}}_{\mathbf{Q}_{1Y\cdot 2}} = \underbrace{\left(\mathbf{Q}_{11} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21}\right)}_{\mathbf{Q}_{11\cdot 2}}\beta_1$$

Step 4. Premultiply by $\mathbf{Q}_{11\cdot 2}^{-1}$:

$$\boxed{\beta_1 = \mathbf{Q}_{11\cdot 2}^{-1}\,\mathbf{Q}_{1Y\cdot 2}}$$

By symmetry: $\beta_2 = \mathbf{Q}_{22\cdot 1}^{-1}\,\mathbf{Q}_{2Y\cdot 1}$.

From Block Elimination to Block Inverse

The block elimination gave us $\beta = \boldsymbol{Q}_{XX}^{-1}\boldsymbol{Q}_{XY}$. What does $\boldsymbol{Q}_{XX}^{-1}$ itself look like?

We want a matrix $\begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{bmatrix}$ such that:

$$\begin{bmatrix} \mathbf{Q}_{11} & \mathbf{Q}_{12} \\ \mathbf{Q}_{21} & \mathbf{Q}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$$

Multiplying out gives four equations. Focus on the first column ($\mathbf{M}_{11}$ and $\mathbf{M}_{21}$):

$$\mathbf{Q}_{11}\mathbf{M}_{11} + \mathbf{Q}_{12}\mathbf{M}_{21} = \mathbf{I} \tag{A}$$

$$\mathbf{Q}_{21}\mathbf{M}_{11} + \mathbf{Q}_{22}\mathbf{M}_{21} = \mathbf{0} \tag{B}$$

Block Inverse: Solving for $\mathbf{M}_{11}$

From (B), solve for $\mathbf{M}_{21}$:

$$\mathbf{M}_{21} = -\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21}\mathbf{M}_{11}$$

Substitute into (A):

$$\begin{aligned}
\mathbf{Q}_{11}\mathbf{M}_{11} + \mathbf{Q}_{12}\left(-\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21}\mathbf{M}_{11}\right) &= \mathbf{I} \\
\left(\mathbf{Q}_{11} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21}\right)\mathbf{M}_{11} &= \mathbf{I} \\
\mathbf{Q}_{11\cdot 2}\,\mathbf{M}_{11} &= \mathbf{I}
\end{aligned}$$

$$\boxed{\mathbf{M}_{11} = \mathbf{Q}_{11\cdot 2}^{-1}}$$

And substituting back: $\mathbf{M}_{21} = -\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21}\mathbf{Q}_{11\cdot 2}^{-1}$.

The same procedure on the second column ($\mathbf{M}_{12}$, $\mathbf{M}_{22}$) gives $\mathbf{M}_{22} = \mathbf{Q}_{22\cdot 1}^{-1}$ and $\mathbf{M}_{12} = -\mathbf{Q}_{11\cdot 2}^{-1}\mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}$.

The Complete Block Inverse

Collecting all four blocks:

$$\boldsymbol{Q}_{XX}^{-1} = \begin{bmatrix} \mathbf{Q}_{11\cdot2}^{-1} & -\mathbf{Q}_{11\cdot2}^{-1}\mathbf{Q}_{12}\mathbf{Q}_{22}^{-1} \\ -\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21}\mathbf{Q}_{11\cdot2}^{-1} & \mathbf{Q}_{22\cdot1}^{-1} \end{bmatrix}$$

Now multiply $\boldsymbol{Q}_{XX}^{-1}\boldsymbol{Q}_{XY}$ to recover our earlier result:

$$\begin{aligned} \beta_1 &= \mathbf{Q}_{11\cdot2}^{-1}\mathbf{Q}_{1Y} - \mathbf{Q}_{11\cdot2}^{-1}\mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{2Y} \\ &= \mathbf{Q}_{11\cdot2}^{-1}\left(\mathbf{Q}_{1Y} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{2Y}\right) = \mathbf{Q}_{11\cdot2}^{-1}\mathbf{Q}_{1Y\cdot2} \quad \checkmark \end{aligned}$$

- We can use $\beta_1 = \mathbf{Q}_{11\cdot2}^{-1}\mathbf{Q}_{1Y\cdot2}$ to understand multivariate regression.

Interpreting Multivariate Regression

- The multivariate projection equation is $Y = X_1'\beta_1 + X_2'\beta_2 + e$.
- Decompose X_1 into the part predicted by X_2 and the part that is not:

$$X_1 = \underbrace{X_2'\gamma_2}_{\text{explained by } X_2} + \underbrace{u_1}_{\text{residual}}$$

- β_1 is the projection of Y onto u_1—the variation in X_1 that is *unique* to X_1, not shared with X_2.
- **Example:** Regress growth on democracy (X_1) and institutions (X_2). β_1 captures the effect of democracy *after removing* what institutions already explain about democracy.

Demonstration: Iterated Projection

Divide X into a single variable in X_1, with all the other variables in X_2, so that

$$Y = X_1'\beta_1 + X_2'\beta_2 + e$$

Regress X_1 on X_2:

$$\begin{aligned}
X_1 &= X_2'\gamma_2 + u_1 \\
\mathbb{E}[X_2 u_1] &= 0 \\
\gamma_2 &= \mathbf{Q}_{22}^{-1}\mathbf{Q}_{21} \\
\mathbb{E}[u_1^2] &= \mathbf{Q}_{11\cdot 2} \\
\mathbb{E}[u_1 Y] &= \mathbf{Q}_{1Y} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{2Y} = \mathbf{Q}_{1Y\cdot 2} \\
\beta_1 &= \mathbf{Q}_{11\cdot 2}^{-1}\mathbf{Q}_{1Y\cdot 2} = \frac{\mathbb{E}[u_1 Y]}{\mathbb{E}[u_1^2]}
\end{aligned}$$

Omitted Variable Bias

- **Long regression**: $Y = X_1'\beta_1 + X_2'\beta_2 + e$. **Short regression**: $Y = X_1'\gamma_1 + u$.
- Substitute the long into the short:

$$\begin{aligned}\gamma_1 &= (\mathbb{E}[X_1X_1'])^{-1}\mathbb{E}[X_1Y] \\ &= (\mathbb{E}[X_1X_1'])^{-1}\mathbb{E}[X_1(X_1'\beta_1 + X_2'\beta_2 + e)] = \beta_1 + \underbrace{(\mathbb{E}[X_1X_1'])^{-1}\mathbb{E}[X_1X_2']}_{\Gamma_{12}}\beta_2\end{aligned}$$

- Γ_{12}: coefficient from regressing X_2 on X_1 (the **association** between included and omitted).
- Bias $= \Gamma_{12}\beta_2$: product of *how correlated* X_1 and X_2 are and *how much* X_2 *matters* for Y.

OVB is Not Solved by Adding Variables

- Suppose $Y = X'\beta + e = X_1'\beta_1 + X_2'\beta_2 + X_3'\beta_3 + e$
- Suppose we cannot measure X_3. Instead, we can choose either:

$$Y = X_1'\gamma_1 + u_1$$

$$Y = X_1'\delta_1 + X_2'\delta_2 + u_2$$

- $\gamma_1 = \beta_1 + \Gamma_{12}\beta_2 + \Gamma_{13}\beta_3$.
- $\delta_1 = \beta_1 + \Gamma_{13 \cdot 2}\beta_3$.
- Which is better depends on the signs and sizes of these terms!

OVB in Practice: Does Democracy Cause Growth?

- Barro (1996) regresses GDP growth on a democracy index:

$$\text{Growth}_i = \beta_1 \text{Democracy}_i + e_i$$

- But countries that democratize also tend to have stronger property rights, higher education, and colonial histories that affect growth.
- The OVB formula tells us:

$$\gamma_1 = \beta_1 + \underbrace{\Gamma_{12}}_{\substack{\text{correlation of}\\ \text{democracy with}\\ \text{institutions}}} \underbrace{\beta_2}_{\substack{\text{effect of}\\ \text{institutions}\\ \text{on growth}}}$$

- If institutions are positively correlated with democracy ($\Gamma_{12} > 0$) and positively affect growth ($\beta_2 > 0$), then γ_1 overstates the effect of democracy.
- Adding controls can help—but only if they reduce the bias term, not increase it.

Where We Stand

Recap: From Population to Practice

1. **CEF** ($m(X) = \mathbb{E}[Y|X]$): Best predictor, but requires the full joint distribution.
2. **BLP** ($\mathcal{P}[Y|X] = X'\beta$): Best *linear* predictor, requiring only means, variances, and covariances.
3. **Partitioned regression**: Each coefficient captures the effect of a variable *after removing* what other variables predict.
4. **OVB**: Omitting a correlated variable biases coefficients; adding variables doesn't always help.

Next: All of this is in the *population*. Starting next lecture, we move to *samples*—using OLS to estimate β and studying when OLS does a good job.